

| QUICK SHORTCUTS IN Arduino™ IDE                                                                                     |                |
|---------------------------------------------------------------------------------------------------------------------|----------------|
| Checks the sketch for programming errors.                                                                           | Verify         |
| Compiles the sketch and uploads it to the Arduino™ board if there are no errors.                                    | Upload         |
| Creates a new sketch.                                                                                               | New            |
| Presents a menu of all the sketches in your 'sketchbook'. Clicking one will open it within the current window.      | Open           |
| Saves the current sketch.                                                                                           | Save           |
| Opens the serial monitor to check for data sent via <code>serial.print()</code> and <code>serial.println()</code> . | Serial Monitor |

5. **Message Window:** On verifying (compiling) the code for possible mistakes in programming, errors are displayed in this part of the IDE window.
6. **Connection Manager:** Shows the model of the board that's connected to your computer and also the COM port via which communication can be established with the Arduino™.

Apart from the regular menu, the Arduino™ IDE has the following additional menu that helps you program the board.

### Arduino™, do this next (Programming)

Now, let's move on to programming, i.e. instructing the processor what to do. Tread lightly here. Giving wrong instructions can sometimes be fatal. The biggest bug can show up due to forgetting to run the code in a loop.

Those among you well aware of software programming should note that there's a slight difference between software and hardware programming. You have one more thing to manage here the pins, the place where all the action happens.

### So how different is hardware programming?

Writing code for hardware devices is slightly different than writing software. It's similar to writing a driver for your hardware, where pin numbers need to be considered.

With hardware programming, you need to take care of a few extra things to make the pins on the microcontroller work.

- ♦ The pins on a microcontroller can be used to either receive data from a device (input) or provide data to the device (output), but not both simultaneously. So the first thing you need to do is set them as input and output pins.